



$$\textcircled{1} \binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k} \quad \text{Pascal}$$

$$\textcircled{2} \sum_{i=k}^n \binom{i}{k} \Downarrow \binom{n+1}{k+1}$$

曲棍球杆

$$\textcircled{3} k \binom{n}{k-1} = n \binom{n-1}{k-1}$$

committee = chairman

$$\textcircled{4} \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

Vandermondt

	order $\times$	order
n 挑 k	$\frac{n!}{(n-k)!}$	$\binom{n}{k}$
not-replace		
replace	n^k	$\binom{n+k-1}{k}$

概率 = 要素、三公理

1. Sample space $S: \{ \dots \}$

2. Event $E \subseteq S$ 是 S 的任一子集

3. Measurement $\mathcal{P}$ $\Omega = 2^S$ (不一定, 见反例!)

$$\mathcal{P}: \Omega \rightarrow \mathbb{R}$$

1. 非负 $\mathcal{P}(E) \geq 0$

概率就是归一的测度

2. 归一 $\mathcal{P}(S) = 1$ $E_i \cap E_j = \emptyset, i \neq j$

3. 可列可加 $E_1, E_2, \dots$ 互不相交 $\Rightarrow \mathcal{P}(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathcal{P}(E_i)$

例 $S = [0, 1)$

$P([a, b]) = b - a$

$x \sim y: x - y \in \mathbb{Q}$

$\mathbb{Q} = \{q_0, q_1, \dots\}$

$V \quad V_0 = V + q_0, V_1 = V + q_1, \dots$

$\forall i, P(V_i) = P(V)$

$[0, 1) = \bigcup_{i=0}^{\infty} V_i = \sum_{i=0}^{\infty} P(V_i) = \sum_{i=0}^{\infty} P(V) < \infty$

不可数集
Vitali集

1. S
2. $\mathcal{F}$ 是 σ -代数
3. $P: \mathcal{F} \rightarrow \mathbb{R}$

1. $S \in \mathcal{F}$
2. $E \in \mathcal{F} \Rightarrow E^c \in \mathcal{F}$
3. 可列并封闭 $E_1, E_2, \dots \in \mathcal{F}$
 $\bigcup_{i=0}^{\infty} E_i \in \mathcal{F}$

$$0 \leq P(E) \leq 1$$

$$P(\emptyset) = 0$$

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i)$$

$$1. P(E) = 1 - P(E^c)$$

365 天, 有人生日在同一天

$$\binom{m}{n} \left(1 - \frac{A_m^n}{m^n}\right)$$

E, F

$$2. E = (E \cap F) \cup (E \cap F^c)$$

$$P(E) = P(E \cap F) + P(E \cap F^c)$$

生日相邻 $n=15$

01010000

$$1 - \frac{1}{m^n} \frac{m}{m-n} \binom{m-n}{n} n! \binom{m-n}{n} \left| \begin{array}{cc} m-1 & n \\ (m-1-n)+1 & (m-3) & n-1 \\ & (m-3)(n-1)+1 \\ & \binom{m-n-1}{n-1} \end{array} \right.$$

$$3. E_1, E_2 \quad P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) \quad (E_1 \cap E_2)$$

$$E_1, E_2, E_3: P(E_1 \cup E_2 \cup E_3) =$$

$$E_1, \dots, E_r: P(E_1 \cup \dots \cup E_r) = \sum_{r=1}^n (-1)^{r+1} \sum_{1 \leq i_1 < \dots < i_r \leq n} P(E_{i_1}, \dots, E_{i_r})$$

Inclusion-Exclusion equation.

4 de Montmort Problem 錯排

A_i : 等 i 个人拿到自己的帽子

$$P(A_1^c \cap A_2^c \dots \cap A_n^c) = 1 - P(A_1 \cup \dots \cup A_n)$$

$$= 1 - \left\{ \begin{aligned} &P(A_1) + \dots + P(A_n) \\ &- P(A_1 A_2) - \dots \\ &+ \dots \end{aligned} \right\} = n \frac{(n-1)!}{n!} = 1$$

$$= \binom{n}{2} \frac{(n-2)!}{n!} = \frac{1}{2!}$$

$$= \binom{n}{3} \frac{(n-3)!}{n!} = \frac{1}{3!}$$

$$\begin{aligned} &= 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots = \sum_{k=0}^n (-1)^k \frac{1}{k!} \\ &= \frac{1}{e} \quad (n \rightarrow \infty) \end{aligned}$$